

# Myasthenia Gravis (MG) in Dogs and Cats

## What is it?

Myasthenia gravis is a disorder of the connection between **nerves and muscles**. The muscle's receptors for the chemical messenger acetylcholine are reduced or damaged, so nerve signals do not reach the muscle well. The result is **weakness that gets worse with activity** and improves with rest.

## Types

- **Acquired MG** (most common) — an **auto-immune** disease where the immune system attacks the receptors. Usually starts at 4 months–4 years or 9–13 years in dogs (2–3 or 9–10 years in cats). More common in German Shepherds, Golden Retrievers, Akitas, terriers, German Shorthaired Pointers, Chihuahuas, and Abyssinian/Somali cats.
- **Congenital MG** — rare, inherited, appears at 6–9 weeks of age (Springer Spaniels, Jack Russell and Smooth Fox Terriers, Samoyeds, Miniature Dachshunds). Miniature Dachshunds may outgrow it by 6 months.

## Signs

- **Focal:** regurgitation from an enlarged esophagus (**megaesophagus**), trouble swallowing, voice change, dropped jaw, weak blink
- **Generalized:** weakness after a few minutes of exercise (often back legs first), tremors, regurgitation; cats may hold the neck bent down
- **Acute fulminating:** rapid, severe weakness, unable to walk, **breathing difficulty** — an emergency with a poor prognosis

## Associated conditions

Thymoma (a chest tumor — 15–26% of cats, ~3% of dogs), hypothyroidism in dogs, other cancers, and methimazole treatment in cats.

## Diagnosis

- **AChR antibody blood test** — confirms acquired MG
- **Chest X-rays** for megaesophagus, pneumonia or a chest mass
- Nerve and muscle function tests, thyroid testing

## Treatment

- **Pyridostigmine** — improves nerve-to-muscle signaling
- **Immunosuppressive drugs** (e.g., prednisone, azathioprine) in some cases
- **Surgery** to remove a thymoma, if present
- **Supportive care:** treat aspiration pneumonia; feeding tube or breathing support if needed

## Feeding a pet with megaesophagus

- Feed in an **upright position** (e.g., a Bailey chair) and keep upright **10–15 minutes** after eating
- Try different food textures (meatballs, slurry) to see what is kept down best

## Medications to avoid

Tell every veterinarian your pet has MG. Avoid when possible: aminoglycoside, tetracycline and ampicillin antibiotics, some heart-rhythm drugs, certain anesthetics and muscle relaxants, and **organophosphate** flea/tick products.

## Prognosis

Guarded in dogs, mainly because of megaesophagus and aspiration pneumonia; 1-year mortality is reported at 40–60%. But **many dogs go into spontaneous remission** — in one study, 47 of 53 dogs on pyridostigmine alone did so in about 6 months. Cats often respond better with immunosuppressive drugs and are more sensitive to pyridostigmine.

## Call us right away if

Coughing, fast or difficult breathing, fever or lethargy (**signs of aspiration pneumonia**), worsening weakness, or frequent regurgitation.